ECE408/CS483/CSE408 Fall 2024

Applied Parallel Programming

Lecture 17
Parallel Sparse Methods

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Course Reminders

- Lab 7 is due this Friday
- Project Milestone 2
  - Due next week
- Take a note of the day/time of midterm 2

Objective

- To learn the key techniques for compacting input data in parallel sparse methods for reduced consumption of memory bandwidth
  - better utilization of on-chip memory
  - fewer bytes transferred to on-chip memory
  - Better utilization of global memory
  - Challenge: retaining regularity

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Sparse Matrix

- Many real-world systems are sparse in nature
  - Linear systems described as sparse matrices
- Solving sparse linear systems
  - Iterative Conjugate Gradient solvers based on sparse matrix-vector multiplication is a common method
- Solution of PDE systems can be formulated into linear operations expressed as sparse matrix-vector multiplication

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Sparse Matrix in Scientific Computing

Science Area	Number of Teams	Codes	Struct Grids	Unstruct Grids	Dense Matrix	Sparse Matrix	N-Body	Monte Carlo	FFT	PIC	Sig I/O
Climate and Weather	3	CESM, GCRM, CM1/WRF, HOMME	X	X		X		X			X
Plasmas/Magnetosphere	2	H3D(M),VPIC, OSIRIS, Magtail/UPIC	X				X		X		X
Stellar Atmospheres and Supernovae	5	PPM, MAESTRO, CASTRO, SEDONA, ChaNGa, MS-FLUKSS	X			X	X	X		X	X
Cosmology	2	Enzo, pGADGET	X			X	X				
Combustion/Turbulence	2	PSDNS, DISTUF	X						X		
General Relativity	2	Cactus, Harm3D, LazEV	X			X					
Molecular Dynamics	4	AMBER, Gromacs, NAMD, LAMMPS				X	X		X		
Quantum Chemistry	2	SIAL, GAMESS, NWChem			X	X	X	X			X
Material Science	3	NEMOS, OMEN, GW, QMCPACK			X	X	X	X			
Earthquakes/Seismology	2	AWP-ODC, HERCULES, PLSQR, SPECFEM3D	X	X			X				X
Quantum Chromo Dynamics	1	Chroma, MILC, USQCD	X		X	X					
Social Networks	1	EPISIMDEMICS									
Evolution	1	Eve									
Engineering/System of Systems	1	GRIPS,Revisit						X			
Computer Science	1			X	X	X			X		X

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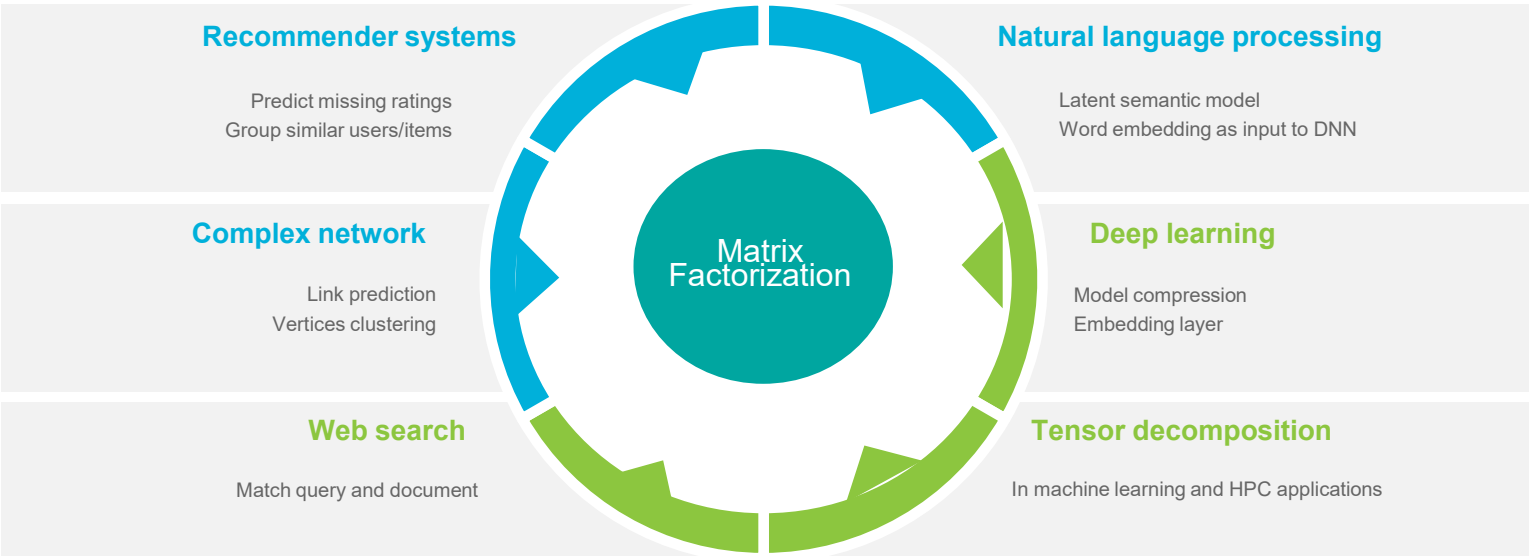
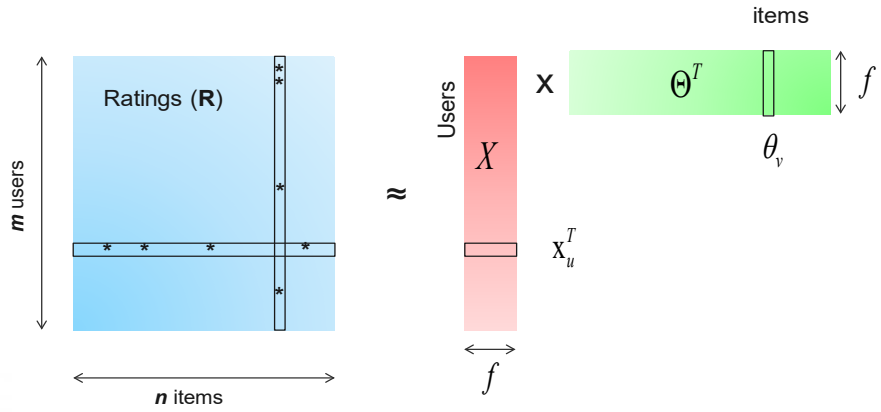
Sparse Matrix in Analytics and AI





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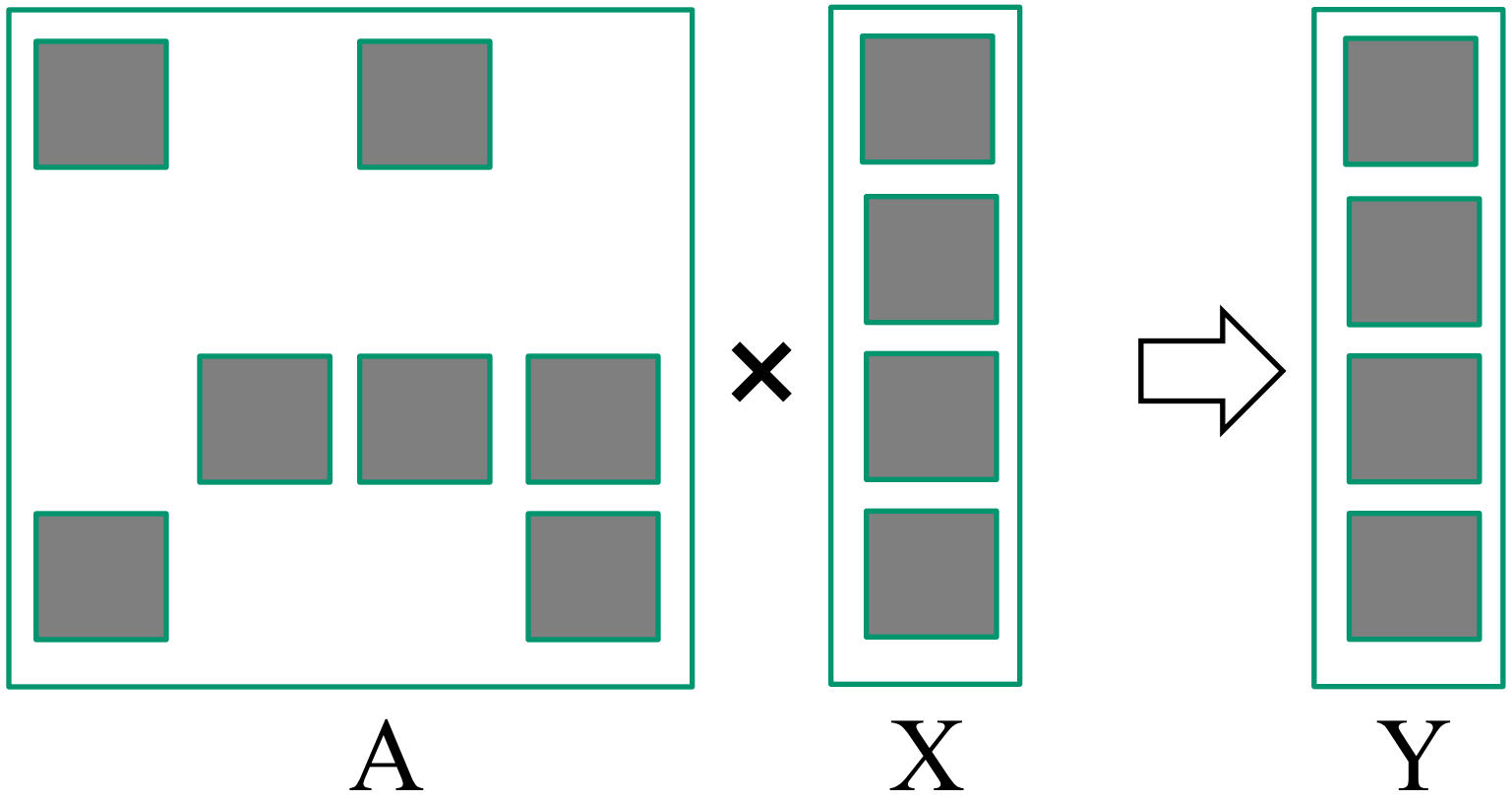
Sparse Matrix-Vector Multiplication (SpMV)



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Challenges

- Compared to dense matrix multiplication, SpMV
  - Is irregular/unstructured
  - Has little input data reuse
  - Benefits little from compiler transformation tools

- Key to maximal performance
  - Maximize regularity (by reducing divergence and load imbalance)
  - Maximize DRAM burst utilization (layout arrangement)

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A Simple Parallel SpMV

Row 0	3	0	1	0	Thread 0
Row 1	0	0	0	0	Thread 1
Row 2	0	2	4	1	Thread 2
Row 3	1	0	0	1	Thread 3

- Each thread processes one row

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Compressed Sparse Row (CSR) Format

CSR Representation

		Row 0	Row 2	Row 3
Nonzero values	`data[7]`	{ 3, 1,	2, 4, 1,	1, 1 }
Column indices	`col_index[7]`	{ 0, 2,	1, 2, 3,	0, 3 }
Row Pointers	`row_ptr[5]`	{ 0, 2,	2, 5, 7 }	

Dense representation

Row 0	3	0	1	0
Row 1	0	0	0	0
Row 2	0	2	4	1
Row 3	1	0	0	1

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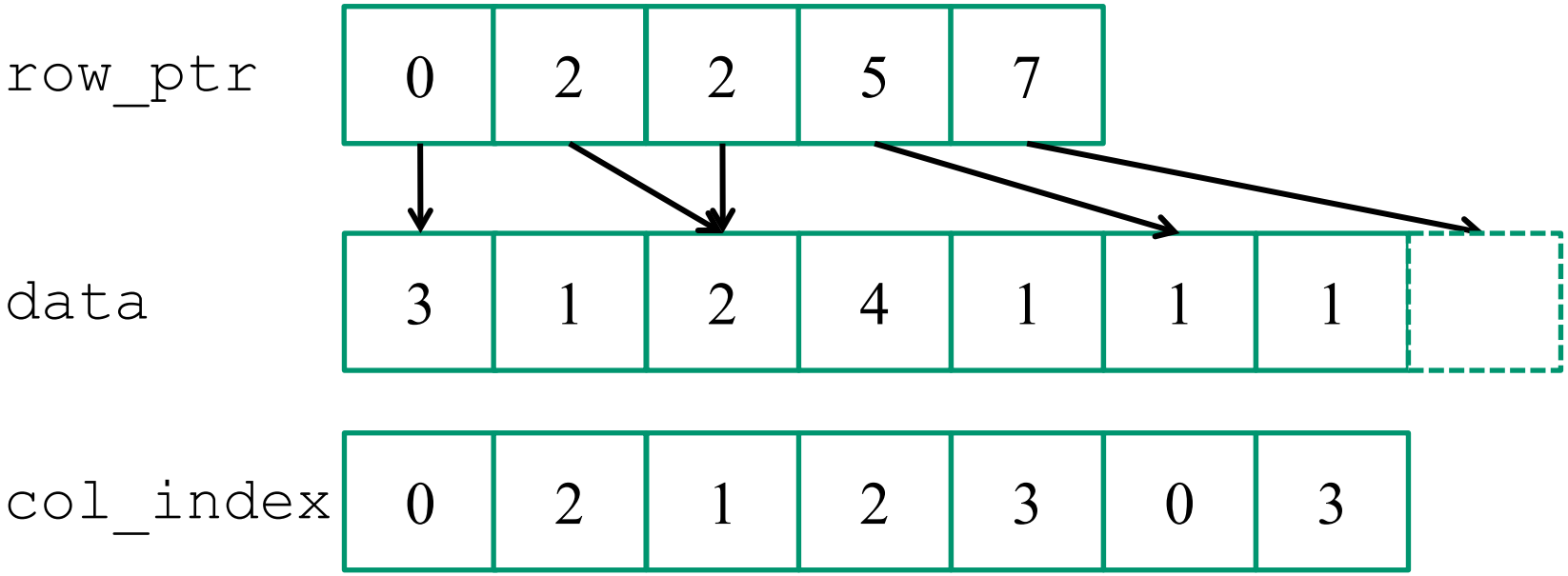
CSR Data Layout



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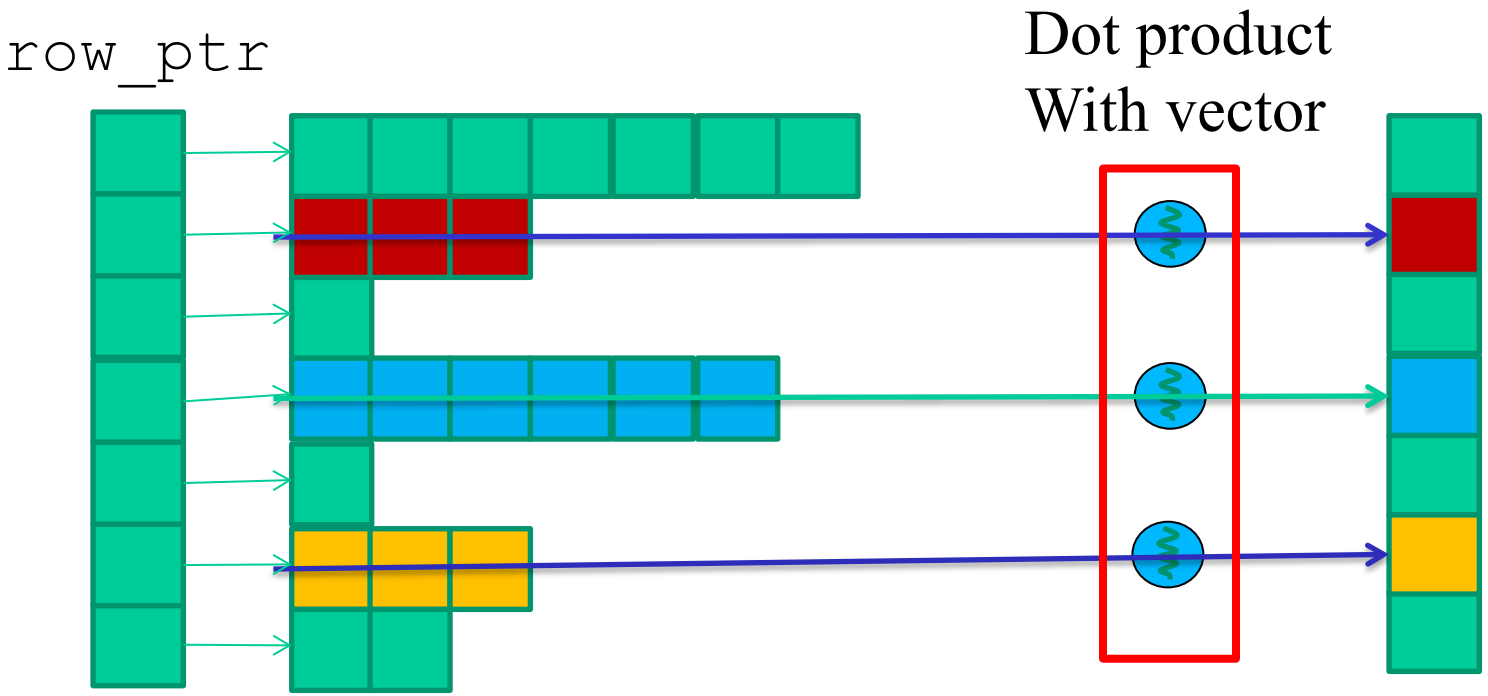
CSR Kernel Design



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A Parallel SpMV/CSR Kernel (CUDA)

```
__global__ void SpMV_CSR(int num_rows, float *data, int
 *col_index, int *row_ptr, float *x, float *y)
{
    int row = blockIdx.x * blockDim.x + threadIdx.x;
    if (row < num_rows) {
      float dot = 0;
      int row_start = row_ptr[row];
      int row_end =   row_ptr[row+1];
      for (int elem = row_start; elem < row_end; elem++)
          dot += data[elem] * x[col_index[elem]];
      y[row] = dot;
    }
}
```

		Row 0	Row 2	Row 3
Nonzero values	`data[7]`	{ 3, 1,	2, 4, 1,	1, 1 }
Column indices	`col_index[7]`	{ 0, 2,	1, 2, 3,	0, 3 }
Row Pointers	`row_ptr[5]`	{ 0, 2,	2, 5, 7 }	

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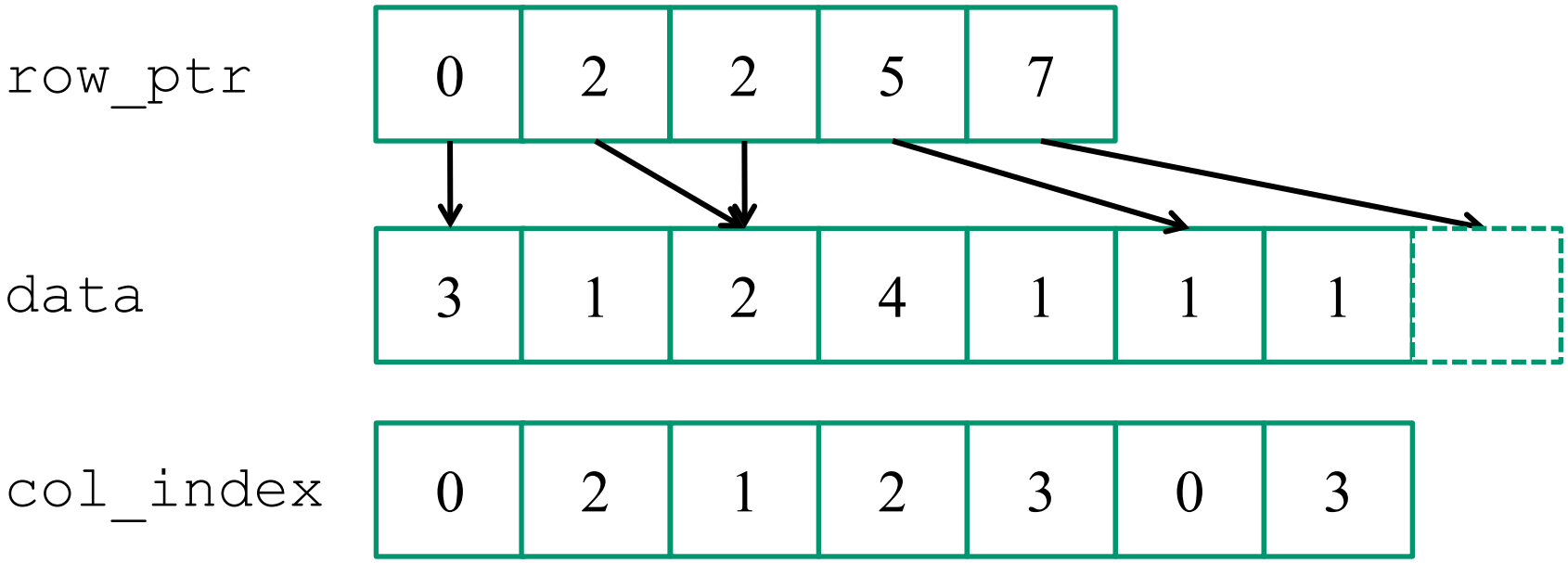
CSR Kernel Control Divergence

- Threads execute different number of iterations



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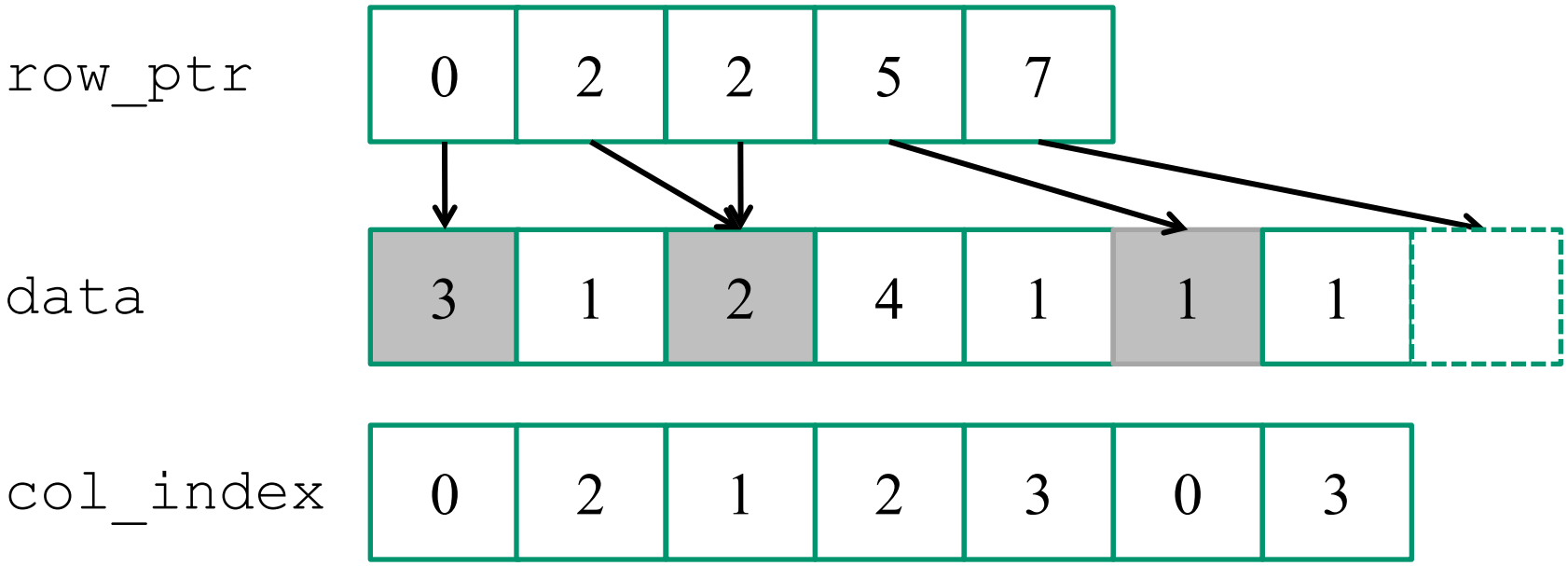
CSR Kernel Memory Divergence (Uncoalesced Accesses)

- Adjacent threads access non-adjacent memory locations
  - Grey elements are accessed by all threads in iteration 0



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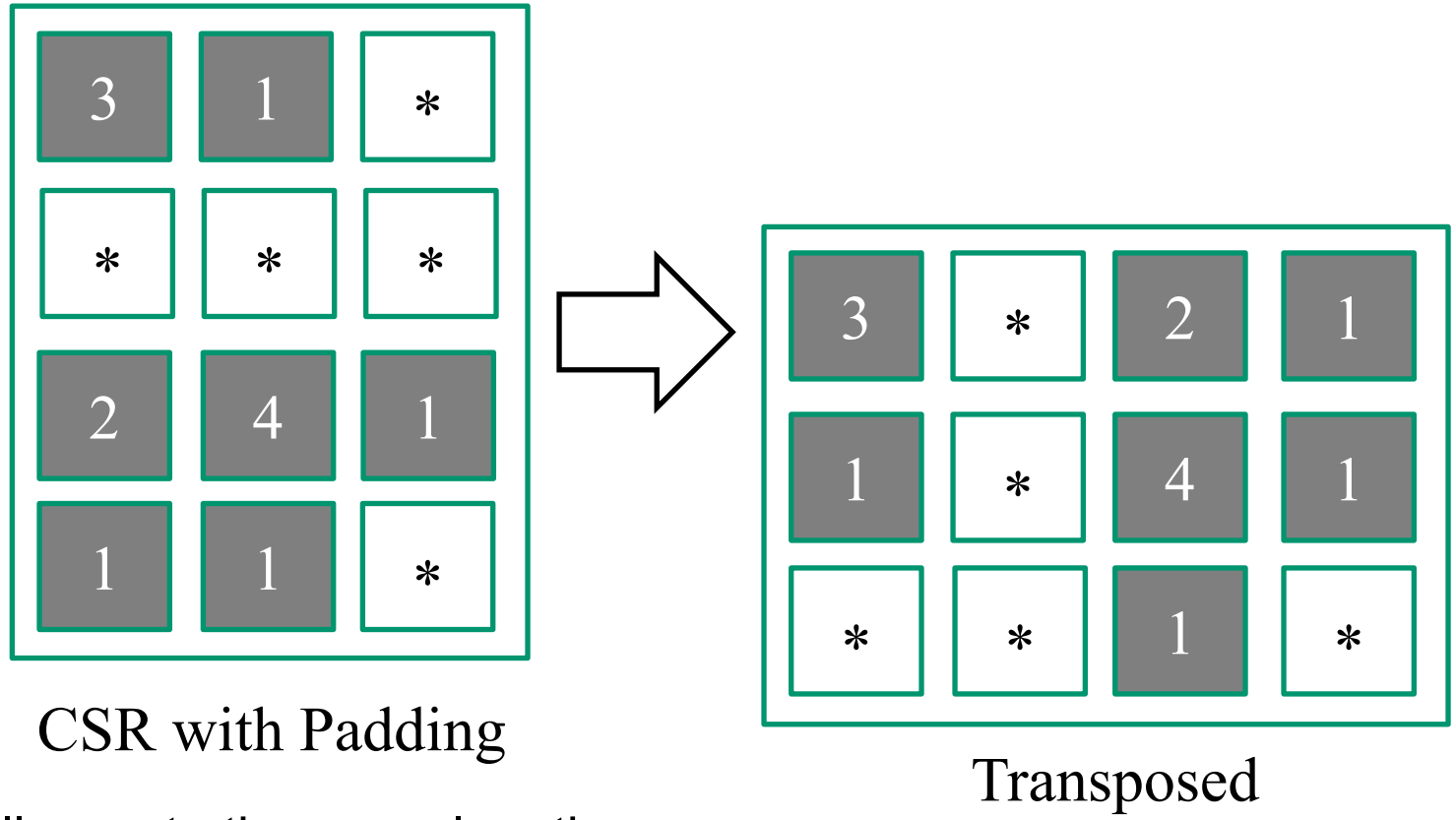
Regularizing SpMV with ELL(PACK) Format



- Pad all rows to the same length
  - Inefficient if a few rows are much longer than others
- Transpose (Column Major) for DRAM efficiency
- Both data and col_index padded/transposed

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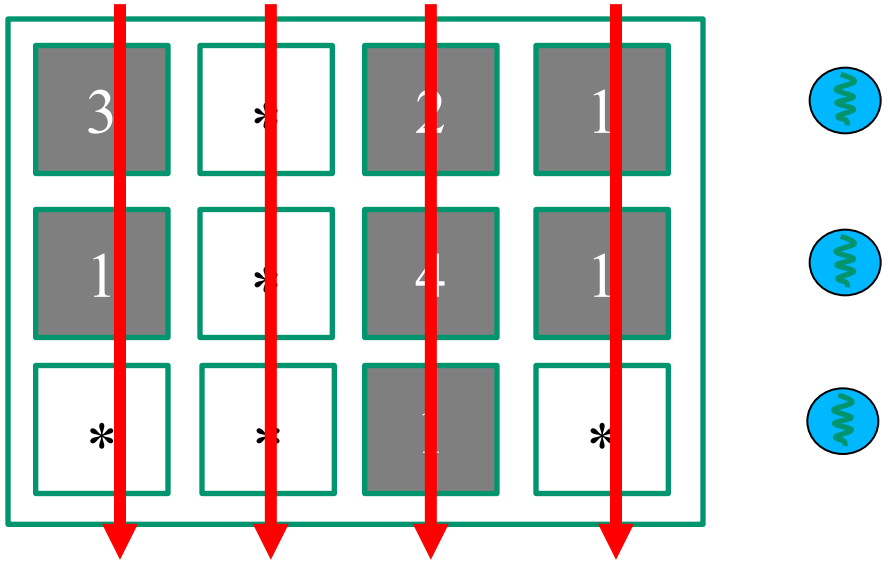
ELL Kernel Design



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A parallel SpMV/ELL kernel

```
__global__ void SpMV_ELL(int num_rows, float *data,
     int *col_index, int num_elem, float *x, float *y)
  {
  int row = blockIdx.x * blockDim.x + threadIdx.x;
  if (row < num_rows) {
    float dot = 0;
    for (int i = 0; i < num_elem; i++)
       dot += data[row+i*num_rows] * x[col_index[row+i*num_rows]];
    y[row] = dot;
   }
  }
```

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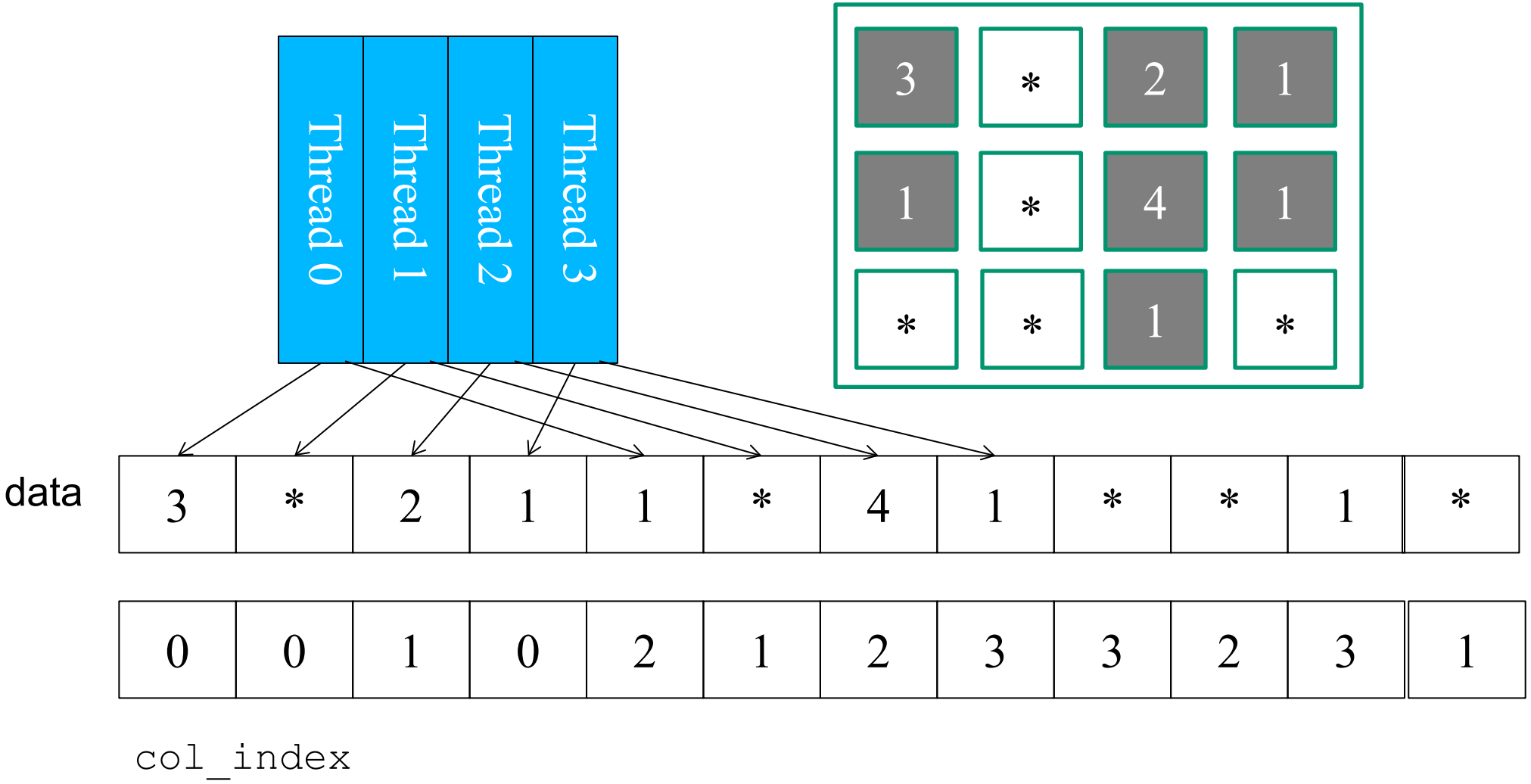
Memory Coalescing with ELL



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Coordinate (COO) format

- Explicitly list the column & row indices for every non-zero element

			Row 0	Row 2	Row 3	
Nonzero values	`data[7]`	{	3, 1,	2, 4, 1,	1, 1	}
Column indices	`col_index[7]`	{	0, 2,	1, 2, 3,	0, 3	}
Row indices	`row_index[7]`	{	0, 0,	2, 2, 2,	3, 3	}

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COO Allows Reordering of Elements

			Row 0	Row 2	Row 3	
Nonzero values	data[7]	{	3, 1,	2, 4, 1,	1, 1	}
Column indices	col_index[7]	{	0, 2,	1, 2, 3,	0, 3	}
Row indices	row_index[7]	{	0, 0,	2, 2, 2,	3, 3	}

Nonzero values	data[7]	{ 1 1, 2, 4, 3, 1 1 }
Column indices	col_index[7]	{ 0 2, 1, 2, 0, 3, 3 }
Row indices	row_index[7]	{ 3 0, 2, 2, 0, 2, 3 }

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COO Kernel

```
for (int i = 0; i < num_elem; i++)
  y[row_index[i]] += data[i] * x[col_index[i]];
```

a sequential loop that implements SpMV/COO

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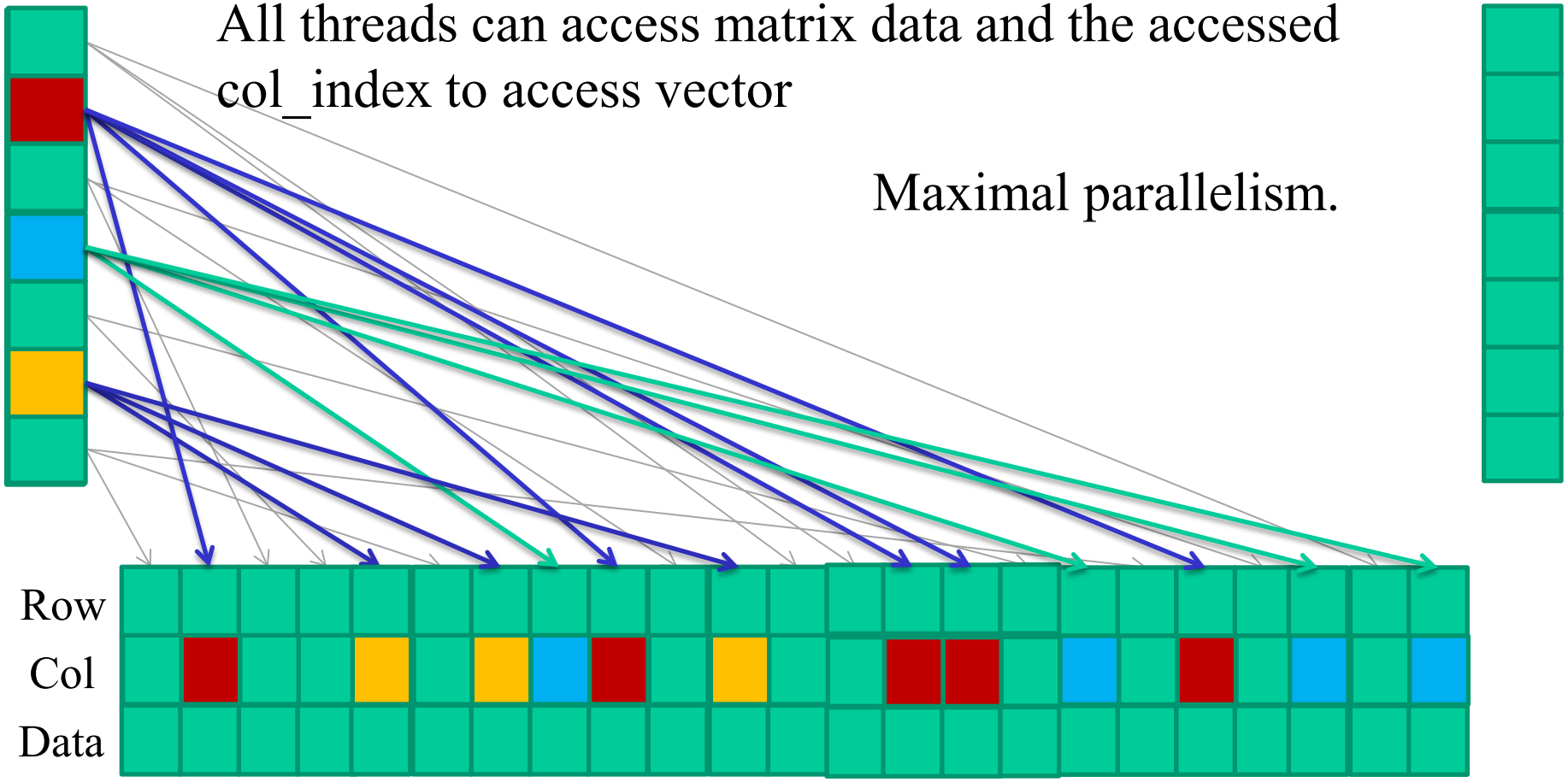
COO Kernel Design
Accessing Input Matrix and Vector

All threads can access matrix data and the accessed col_index to access vector

Maximal parallelism.



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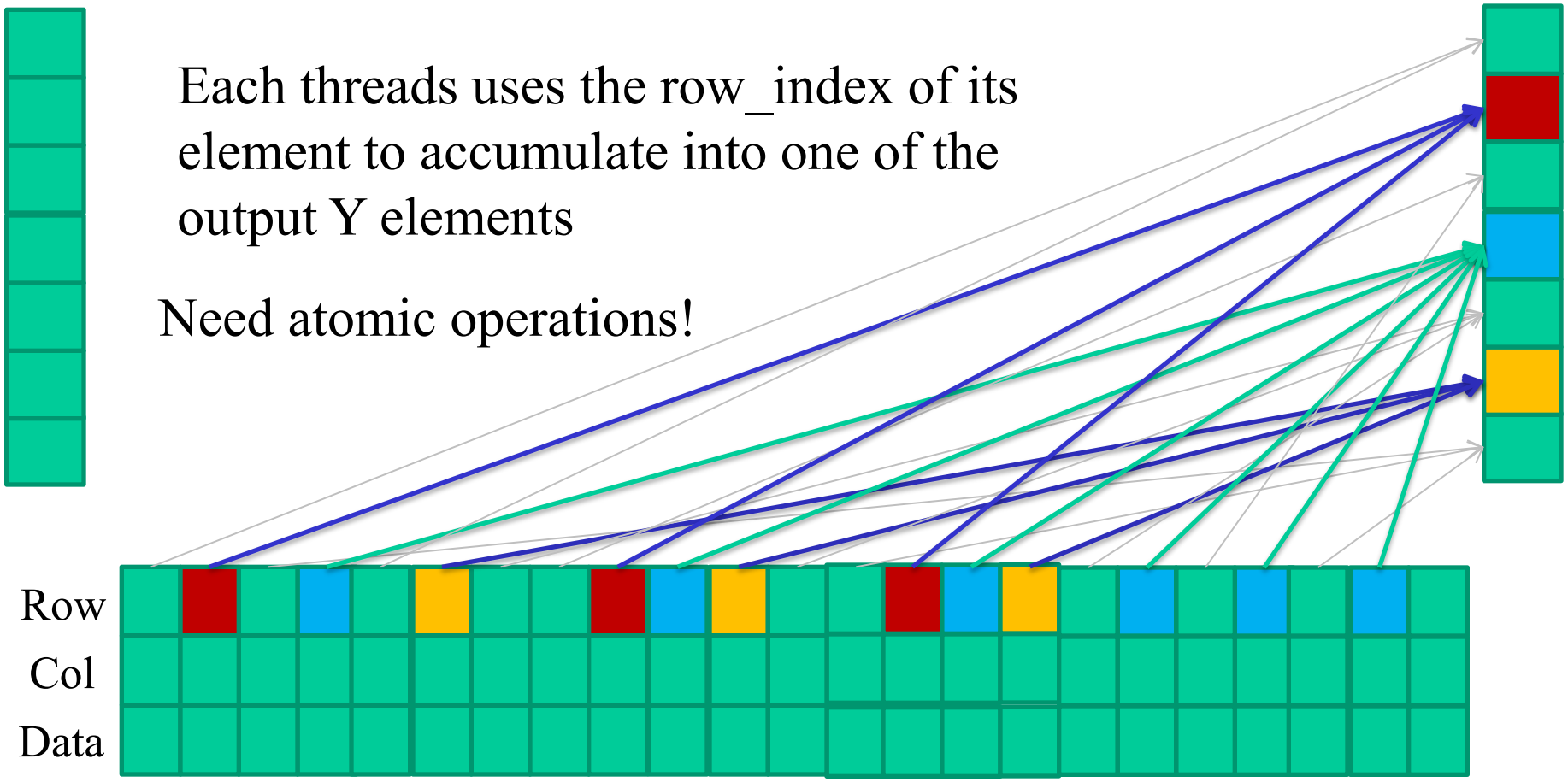
COO kernel Design
Accumulating into Output Vector

Each threads uses the row_index of its element to accumulate into one of the output Y elements

Need atomic operations!



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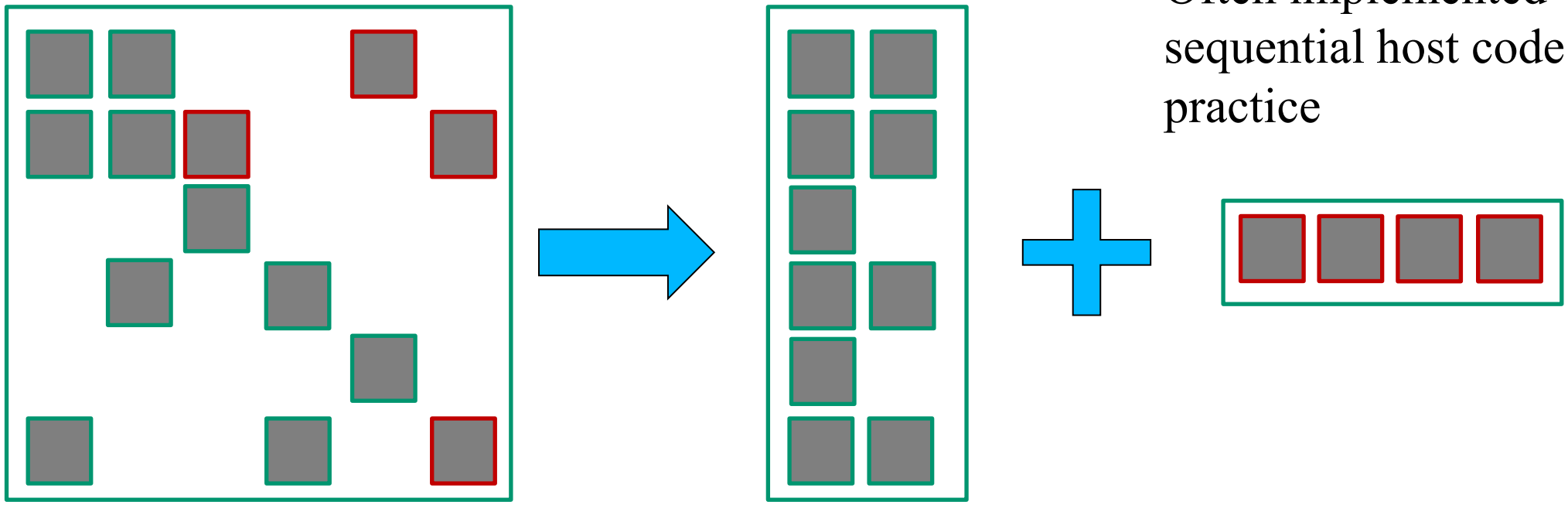
Hybrid Format (ELL + COO)

- ELL handles *typical* entries
- COO handles *exceptional* entries
  - Implemented with segmented reduction



Often implemented in sequential host code in practice

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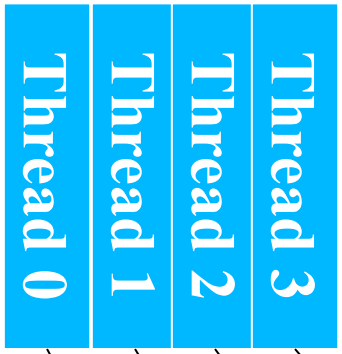
Reduced Padding with Hybrid Format



	data		col_index	
Thread 0	3	1	0	2
Thread 1	*	*	*	*
Thread 2	2	4	1	2
Thread 3	1	1	0	3

Iteration 0

data	3	*	2	1	1	*	4	1
index	0	*	1	0	2	*	2	3

ELL

data	1
col_index	3
row_index	2

COO

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ANY MORE QUESTIONS READ CHAPTER 10

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Problem Solving

- Q: Given matrix A,
  which of the following are correct?

$$A = \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 \\ 1 & 2 & 3 & 4 \\ 1 & 0 & 0 & 1 \end{vmatrix}$$

CSR representation:

Data = [1,2,1,1,2,3,4,1,1]

Col_idx = [0,2,3,0,1,2,3,0,3]

Row_ptr = [0,1,3,7,9]

COO representation

Data = [1,2,1,1,2,3,4,1,1]

Col_idx = [0,2,3,0,1,2,3,0,3]

Row_idx = [0,1,1,3,3,3,3,7,7]

- A: only CSR

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